



Revamping The Technical Education System

"The study of the scientific methods and principles underlying the practice of any handicraft, industry or profession; and the application of those methods and principles of the handicraft, industry, or profession in question. The first is the primary or technological aspect of the subject; the second is its subsequent and practical application."

— 1901, Lord Curzon, the then Viceroy of India.

India produces over eight lakh graduates and diplomas in professional degrees, from over four thousand approved institutes. Yet, there is an acute shortage of the right manpower across industries. A severe attrition problem exists in many industries other than IT; from the candidate's side, there is the ever-looming prospect of unemployment and under-

Uniformity in technical education and not the English language alone will help India win the race to become a global destination for bagging contracts

employment. Typically, the graduation certificate needs an additional certificate that students get independently of their college education—like in IT or management—to find the right job. With campus interviews—the trend not just in the IIMs and the IITs, but even in other universities—the gap between the industrial HR needs and the qualification of the passouts has come to the fore.

Typically, soft skills are most conspicuous in their absence. The stress now is more and more on the need for emotional quotient rather than just the IQ. R Guhan, Country Manager, JobsDB.com says, "Communication and commitment are problems that most companies face while recruiting. The salary expectations are high. And, employees treat jobs that

pay less, a stop-gap arrangement before jumping to greener pastures. In fact, even lecturers and principals are jumping to industrial jobs as they are better paying."

An HR team confirmed this view in a presentation that it made to its management. The company is a leading automotive components maker in Chennai. Many candidates recruited from leading institutions stay for an year or two, pick up the tricks of the trade and move on to more lucrative jobs.

Picking students from the not-so-top class institutes is a dicey affair as there is the ever-present question raised about the quality of education imparted. The well-known capitation fee structure does nothing to dispel the misgivings.

Down Memory Lane

Technical education in India has a rich history, starting with the first survey school in 1794, near Fort St George in Chennai, with eight students. Now it's well known as the Guindy Engineering College. A modest beginning, but by no way insignificant, and by independence, there were over 12,000 students admitted to the engineering colleges.

The Technical Education of Central Advisory Board of Education, 1943, envisaged: "The primary function of technical instruction remains, and is likely to remain, that of satisfying the needs of industry and commerce for a) skilled craftsmen, b) intelligent foremen and executives, and c) research workers." It includes engineering, pharma, applied arts, MBA, MCA and similar professional courses.

Post-independence, the first breakthrough in education happened with the setting up of the Indian Institutes of Technology (IIT), conceived by Prime Minister, Pt Jawaharlal Nehru, and set up in 1950 in Calcutta—later shifted to Kharagpur—and called the Eastern Higher Technical Institute. It was called IIT just before the inauguration in 1951. The other IITs were in Bombay (1958), Madras (1959), Kanpur (1960), Delhi (1963) and Guwahati (1995). The first management institute came around this time, in 1954, and was called the Indian

Meeting The Chinese Challenge

This is a reality that cannot be ignored. China has improved its infrastructure and lifestyle to meet the global needs. In a recent visit to India, the Chinese premier visited the IITs and IISc. "No doubt, this was to emulate the education system followed by these institutions. And what India is doing in small numbers, China will replicate in hundreds. We have but five years before the Chinese catch up," said Kiran Karnik, President, NASSCOM. "Unless the education system in India is revamped dramatically, India cannot retain the advantage it has today," he added, voicing the concern of many.

Institute of Social Welfare and Business Management, in Calcutta.

Private institutions

Post liberalisation, education was freed and private institutions started coming up, especially in Maharashtra, Tamil Nadu, Karnataka and Andhra Pradesh. Even today, these states easily take the lead in the number of institutions and the number of students graduating each year. While initially the purpose of these institutions was envisaged as charitable, today profit making is recognised as the goal of the private-run institutions, which has also thrown up questions of the quality of education imparted.

Technical Education Administration

One of the problems faced by the industry today is the lack of uniformity in technical education, as different institutions are run differently. There are the institutions run by the central and state governments, the universities, the deemed-to-be-universities and private institutions.

The Central government runs its institutions through the Bureau of Technical Education under the Ministry for Human Resources, the UGC and the AICTE. It is responsible for policy formulation, funding AICTE, University Grants Commission (UGC) and national institutes, partial funding of RECs, training and paying stipends to graduates/diploma holders under the Apprenticeship Act, to teachers for training in industry, linking institutions with foreign agencies for developmental activities, etc.

The UGC allocates funds for maintenance and development to universities.

AICTE, instituted in 1945 as an advisory body, takes care of planning, formulation and maintenance of norms and standards, accreditation, funding of priority areas, monitoring and evaluation of institutions, maintaining parity of certification and awards and ensuring the coordinated and integrated development of technical and management education. AICTE looks at the development needs, coordinates development, allocates and disburses funds for technical education, promotes research, etc. It also accredits colleges through the National Bureau of Accreditation (NBA).

The state government runs its technical institutes through the Directorate of Technical Education (DTE) and State Board of Technical Education. While the DTE is responsible for development of technical education, manpower forecast, curriculum design, starting, new institutions, budgeting and human resource development, the State Board undertakes policy planning, formulations and approval of curriculum,